

Topic #1 Assignment

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Theoretical Questions

1. Let:

```
data_a <- c(6, 7, 9, 1, 2, 3, 8, 4, 6)
data_b <- c(10, 9, 8, 7, 5, 4, 1, 7, 6)
data_c <- c(1, 0, 2, 0, 1, 0, 4, 0, 2)

matrix_a <- matrix(data_a, nrow=3, ncol=3, byrow=TRUE)
matrix_b <- matrix(data_b, nrow=3, ncol=3, byrow=TRUE)
matrix_c <- matrix(data_c, nrow=3, ncol=3, byrow=TRUE)
```

i. $3B^T + A$

```
question_i <- 3 * t(matrix_b) + matrix_a

print(question_i)
```

	[,1]	[,2]	[,3]
[1,]	36	28	12
[2,]	28	17	24
[3,]	32	16	24

ii. $C^T - 4A$

```
question_ii <- t(matrix_c) - (4 * matrix_a)

print(question_ii)
```

	[,1]	[,2]	[,3]
[1,]	-23	-28	-32
[2,]	-4	-7	-12
[3,]	-30	-16	-22

iii. $(CA)'$

```
question_iii <- t(matrix_c %*% matrix_a)
```

```
print(question_iii)
```

```
      [,1] [,2] [,3]
[1,]    22     1    40
[2,]    15     2    36
[3,]    21     3    48
```

```
question_iii_2 <- t(matrix_a) %*% t(matrix_c)
```

```
print(question_iii_2)
```

```
      [,1] [,2] [,3]
[1,]    22     1    40
[2,]    15     2    36
[3,]    21     3    48
```

2. Let:

```
X <- matrix(c(1, 2, 3, 4, 5, 6, 7, 8, 9), nrow=3, ncol=3)
beta <- c(10, 20, 30)
```

Verify the following two are equivalent: $X\beta$ and $X_{\cdot,1}\beta_1 + X_{\cdot,2}\beta_2 + X_{\cdot,3}\beta_3$

```
ver1 <- X %*% beta
ver2 <- X[, 1] * beta[1] + X[, 2] * beta[2] + X[, 3] * beta[3]
```

```
print(ver1)
```

```
      [,1]
[1,]   300
[2,]   360
[3,]   420
```

```
print(ver2)
```

```
[1] 300 360 420
```

3. Match each matrix with its inverse:

```
A <- matrix(c(1, 2, -1, -2, 0, 1, 1, -1, 0), nrow=3, ncol=3, byrow=TRUE)
print(A)
```

```
      [,1] [,2] [,3]
[1,]     1     2    -1
[2,]    -2     0     1
[3,]     1    -1     0
```

```
A_inv <- solve(A)
print(A_inv)
```

```
      [,1] [,2] [,3]
[1,]     1     1     2
[2,]     1     1     1
[3,]     2     3     4
```

```
# the inverse of A is b
```

```
B <- matrix(c(2, 4, 1, -1, 1, -1, 1, 4, 0), nrow=3, ncol=3, byrow=TRUE)
print(B)
```

```
      [,1] [,2] [,3]
[1,]     2     4     1
[2,]    -1     1    -1
[3,]     1     4     0
```

```
B_inv <- solve(B)
print(B_inv)
```

```
      [,1] [,2] [,3]
[1,]    -4    -4     5
[2,]     1     1    -1
[3,]     5     4    -6
```

```
# the inverse of B is a
```

```
C <- matrix(c(1, 2, 3, 2, 4, 0, 0, 0, 3), nrow=3, ncol=3, byrow=TRUE)
print(C)
```

```
      [,1] [,2] [,3]
[1,]    1    2    3
[2,]    2    4    0
[3,]    0    0    3
```

```
# C_inv <- solve(C)
print(det(C))
```

```
[1] 0
```

```
# the determinate of C is 0, so the matrix is not invertible
```

4. The distribution of $\begin{bmatrix} 1 & 1 \end{bmatrix} \varepsilon$ is derived as:

$$\begin{bmatrix} 1 & 1 \end{bmatrix} \varepsilon \sim \mathcal{N}(a' \mu, a' \Sigma a)$$

where:

$$a' \mu = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 2$$

$$a' \Sigma a = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \sigma_1^2 + \sigma_2^2 + 2\sigma_{12}$$

Therefore:

$$\begin{bmatrix} 1 & 1 \end{bmatrix} \varepsilon \sim \mathcal{N}(2, \sigma_1^2 + \sigma_2^2 + 2\sigma_{12})$$

5. Let $f(x, y) = x^2 + y^2 + 2xy$

i. What is $\frac{\partial f}{\partial x} f(x, y)$ and $\frac{\partial f}{\partial y} f(x, y)$?

$$\frac{\partial f}{\partial x} = 2x + 2y$$

$$\frac{\partial f}{\partial y} = 2y + 2x$$

ii. Use this to create a Taylor approximation of this function at (x_0, y_0) . Write this in the form of:

$$f(x_0 + dx, y_0 + dy) \approx f(x_0, y_0) + \begin{bmatrix} 2x_0 + 2y_0 & 2y_0 + 2x_0 \end{bmatrix} \begin{bmatrix} dx \\ dy \end{bmatrix}$$

$$f(x_0 + dx, y_0 + dy) \approx f(x_0, y_0) + \begin{bmatrix} \frac{\partial f}{\partial x}(x_0, y_0) & \frac{\partial f}{\partial y}(x_0, y_0) \end{bmatrix} \begin{bmatrix} dx \\ dy \end{bmatrix}$$

$$= f(x_0, y_0) + \begin{bmatrix} 2x_0 + 2y_0 & 2y_0 + 2x_0 \end{bmatrix} \begin{bmatrix} dx \\ dy \end{bmatrix}$$

$$= x_0^2 + y_0^2 + 2x_0y_0 + (2x_0 + 2y_0)dx + (2y_0 + 2x_0)dy$$

iii. Plug in $(x_0, y_0) = (0, 0)$ to create a linear approximation to this function

At $(x_0, y_0) = (0, 0)$:

$$f(0 + dx, 0 + dy) \approx 0 + \begin{bmatrix} 0 & 0 \end{bmatrix} \begin{bmatrix} dx \\ dy \end{bmatrix} = 0$$